

JUAN NICOLÁS MENDOZA RONCANCIO

Data and Artificial Intelligence Engineering Student at Mines Paris - PSL

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Profile

Msc in Engineering Student at Mines Paris - PSL, specializing in **AI, machine learning and deep learning**. Experienced in **Python, SQL, PyTorch, Git** with experience in **AI model deployment**. Recognized for **teamwork** and **adaptability**. Seeking a **4-6 month international internship** starting **May 2026** in **data and artificial intelligence** in France or internationally.

Education

Mines Paris - PSL

MSc in Engineering Major in applied mathematics, Data and Artificial Intelligence

2025 – Present

Paris, France

National University of Colombia

Bachelor's in Mathematics

2020 – Present

Bogotá, Colombia

- **Top of class**
- Academic exchange in **Mathematics and AI** at University of los Andes

National University of Colombia

Certified programs in AI and Data Science

2023, 2024

Bogotá, Colombia

- Developing applications with **Large Language Models**
- Advanced **Machine Learning** and Data Science
- Machine Learning and **Data Science**
- Summer school in **High Performance Computing** for **Big Data** applications

Experience

Research Assistant – Convex Clustering

Mathematics Department, University of los Andes

March 2025 – Present

Bogotá, Colombia

- Implementation of **Linear Boosting algorithms** applied to **data clustering**.
- Achieved competitive metrics: **Silhouette Coefficient 0.5** and **BDI 1.25**, comparable to classical methods such as **KNN** or **SVM**.

Research Assistant

MindLab Research Group

March 2024 – Present

Bogotá, Colombia

- Development in **Python** of **medical image classification models** using **KDM** achieving metrics such as **accuracy of 0.95**, comparable to standard CNNs.
- Deployment on **cloud servers** with connection to **Hugging Face** for medical image classification experiments **using LLMs**, achieving 0.98 accuracy with models such as **MedGemma**.

Research Assistant, Underwater Network Dynamics Project

Austral University of Chile & National University of Colombia

September 2024 – February 2025

Chile / Colombia

- Computational modeling and implementation in **Julia** of a model based on **mass-spring systems** of underwater networks based on their interaction with water depending on density and depth.

Research Assistant – Generative Models

National University of Colombia

March 2024 – August 2025

Bogotá, Colombia

- Design of a **BitNet-type generative model** for producing **educational resources and study guides**.

Data Analyst

National University of Colombia

March 2022 – October 2022

Bogotá, Colombia

- Collection and analysis of data on over **500 high-consumption household appliances**, identifying key energy efficiency trends.
- Data cleaning and preprocessing, achieving a **95% improvement rate**.
- Development of decision criteria and presentation of recommendations, enabling **20% improvement in energy efficiency** and **15% reduction in energy consumption**.

Projects

ChatBot on Colombian Armed Conflict | *Python, Keras, Pandas, PyTorch*

January 2024

- Development of a **ChatBot** trained on **940 testimonies** from the Colombian armed conflict.
- Design of **neural network architectures (SimpleRNN, LSTM, GRU)** for natural language processing.
- Application of **language preprocessing techniques** and **hyperparameter optimization** to improve performance.

Sentiment Analysis on Climate Change | *Python*

July 2023

- Analysis of a corpus of **9050 tweets** related to climate change, collected directly from Twitter.
- Conducted **sentiment analysis** using LDA algorithm and **hyperparameter search**, identifying a total of **6 topics**.
- Evaluation of identified sentiments to **find correlations**, obtaining minimum correlation of $r^2 = 0.589$ and maximum of $r^2 = 0.8941$.

Skills

Programming Languages: Python, C++, Matlab, Julia

Frameworks & Libraries: PyTorch, Hugging Face, Keras, Scikit-Learn.

Databases: SQL, MongoDB, Cassandra.

Soft Skills: Teamwork, critical thinking, adaptability, academic leadership

Languages: English C1, French B2, Native Spanish.